Question 2

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Do students who have taken both 131 and 134 report greater interest in machine-learning-related fields than students who have taken 131 only?

Our second question surrounds the association between taking certain PSTAT courses and having specific areas of interest. Specifically, we want to see if taking PSTAT 131 and/or PSTAT 134 has any correlation in machine learning domain interests. We chose the 131/134 courses because its course description fits the topic of machine learning and its techniques pretty well. It is a course that is well known in the PSTAT department, and with the ongoing advancements in artificial intelligence and machine learning, we only feel that the latter is worth exploring through our survey data.

```
problem2 <- read_csv(here::here("data", "processed", "q2", "problem2.csv"))
```

```
## Rows: 49 Columns: 15
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (11): prog.prof, math.prof, stat.prof, upper_div_courses, courses, domai...
## dbl  (4): response_id, prog.comf, math.comf, stat.comf
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
dim(problem2)
```

```
## [1] 49 15
```

```
head(problem2)
```

```
## # A tibble: 6 x 15
##   response_id prog.prof prog.comf math.prof math.comf stat.prof stat.comf
##         <dbl> <chr>         <dbl> <chr>         <dbl> <chr>         <dbl>
## 1        1001 int               4 int               4 adv               5
## 2        1002 int               4 int               3 int               5
## 3        1003 beg               3 int               3 adv               4
## 4        1004 adv               4 adv               5 adv               5
## 5        1005 adv               4 int               4 adv               5
## 6        1006 int               4 int               2 adv               4
## # i 8 more variables: upper_div_courses <chr>, courses <chr>,
## #   domain_specialization <chr>, research <chr>, type <chr>, language <chr>,
## #   domains <chr>, areas <chr>
```

We have already joined the two datasets (**background-clean.csv** and **interest-clean.csv**) by `response_id`. We can see that there are 49 respondents who consented to sharing both responses. While there are 15 columns, we will mainly be using `courses` and `areas`.

Data Validation

We are assuming that students who have taken PSTAT 134 have also taken PSTAT 131 since it is a pre-requisite to that course. We checked this assumption by ensuring that every student who listed PSTAT 134 also included PSTAT 131.

```
course_check <- problem2 %>%
  mutate(
    took_131 = str_detect(courses, "\\bPSTAT 131\\b"),
    took_134 = str_detect(courses, "\\bPSTAT 134\\b")
  )

# Show everyone who took PSTAT 134 but not PSTAT 131
course_check %>%
  filter(took_134 & !took_131) %>%
  select(response_id, courses)
```

```
## # A tibble: 0 x 2
## # i 2 variables: response_id <dbl>, courses <chr>
```

Since there are no students who took PSTAT 134, but not PSTAT 131, our data is valid.

Now let's see the potential options for area interests:

```
possible_values <- unique(unlist(strsplit(as.character(problem2$areas), split = ",")))

possible_values
```

```
##  [1] "Spatial statistics or time series analysis"
##  [2] " Data acquisition and engineering"
##  [3] " Data visualization and interactive dashboards"
##  [4] " Predictive modeling"
##  [5] " generally"
##  [6] " Statistical models and inference"
##  [7] "Analysis or classification of images"
##  [8] " Spatial statistics or time series analysis"
##  [9] " Natural language processing and analysis of text"
## [10] " Model deployment and software or web integrations"
## [11] " Databases"
## [12] " Algorithms"
## [13] " Deep learning and neural networks"
## [14] "Deep learning and neural networks"
## [15] "Data acquisition and engineering"
## [16] "Model deployment and software or web integrations"
```

Out of the eleven options (not counting spacing discrepancies), we will define 'machine learning' interests as one of the following six options:

```
ml_areas <- c(
  "Analysis or classification of images",
  "Deep learning and neural networks",
  "Data acquisition and engineering",
  "Natural language processing and analysis of text",
  "Predictive modeling, generally",
  "Statistical models and inference, generally"
)
```

We chose these six areas because they were the most closely related to machine learning. Students were able to select from eleven areas of interest, and most of them can be argued to connect to ML in some way. However, this selection focuses primarily on areas involving building, training, or applying predictive models, or preparing the data those models rely on. We recognize that this choice of selection is somewhat subjective and that a different selection could lead to slightly different results.

PSTAT 131 vs non-PSTAT 131

To begin, we will explore students who have taken PSTAT 131 and students who have not taken PSTAT 131.

```
enrollment_table <- problem2 |>
  mutate(
    course_group = if_else(
      str_detect(coalesce(courses, ""), "\\bPSTAT 131\\b"),
      "Took PSTAT 131",
      "Did not take PSTAT 131"
    )
  ) %>%
  count(course_group, name = "Number of Students") %>%
  mutate(
    Proportion = `Number of Students` / sum(`Number of Students`)
  )

save_table(enrollment_table, "pstat131_enrollment.csv")

kable(
  enrollment_table,
  col.names = c(
    "PSTAT 131 Enrollment",
    "Number of Students",
    "Proportion"
  ),
  digits = 3,
  caption = "Students by PSTAT 131 Enrollment"
)
```

Table 1: Students by PSTAT 131 Enrollment

PSTAT 131 Enrollment	Number of Students	Proportion
Did not take PSTAT 131	5	0.102
Took PSTAT 131	44	0.898

Here, we can see that only 5 students have not taken PSTAT 131, totaling around 10% of the respondents. On the other hand, 90% of students have taken PSTAT 131 at 44 students.

```
ml_data_131 <- problem2 |>
  mutate(course_group = if_else(str_detect(courses, "\\bPSTAT 131\\b"),
                                "Took PSTAT 131", "Did not take PSTAT 131")) |>
  crossing(area = ml_areas) |>
  mutate(selected = str_detect(areas, fixed(area))) |>
  group_by(response_id, course_group) |>
  summarise(n_ml_areas = sum(selected), .groups = "drop")

ml_data_131
```

```
## # A tibble: 49 x 3
##    response_id course_group           n_ml_areas
##          <dbl> <chr>                       <int>
##  1        1001 Took PSTAT 131                  3
##  2        1002 Did not take PSTAT 131          5
##  3        1003 Took PSTAT 131                  6
##  4        1004 Took PSTAT 131                  2
##  5        1005 Took PSTAT 131                  6
##  6        1006 Took PSTAT 131                  3
##  7        1007 Took PSTAT 131                  6
##  8        1008 Took PSTAT 131                  5
##  9        1009 Took PSTAT 131                  5
## 10        1010 Took PSTAT 131                  5
## # i 39 more rows
```

Now we have added another component: the number of machine learning area interests each student selected. We have printed out the raw table to see this addition.

```
ml_summary <- ml_data_131 %>%
  group_by(course_group) %>%
  summarise(
    n = n(),
    mean_ml_areas = mean(n_ml_areas),
    median_ml_areas = median(n_ml_areas),
    .groups = "drop"
  )

save_table(ml_summary, "ml_interest_summary_by_pstat131.csv")

kable(
  ml_summary,
  digits = 3,
  caption = "ML-Related Interests by PSTAT 131 Enrollment"
)
```

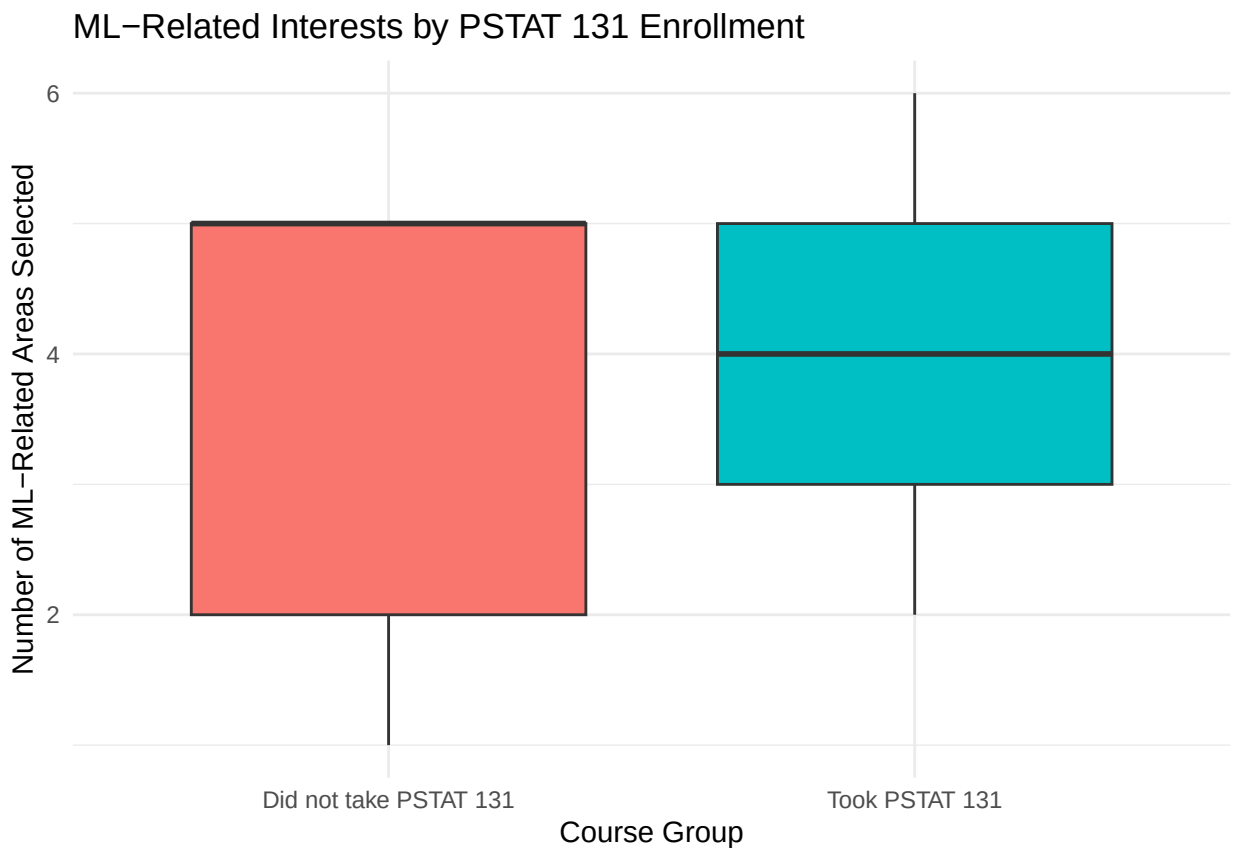
Table 2: ML-Related Interests by PSTAT 131 Enrollment

course_group	n	mean_ml_areas	median_ml_areas
Did not take PSTAT 131	5	3.6	5
Took PSTAT 131	44	4.0	4

Here we have summary statistics for students who have taken PSTAT 131 vs students who have not taken PSTAT 131. The mean and median number of machine learning areas are shown. The mean is larger for students who took PSTAT 131, while the median is larger for students who have not taken PSTAT 131.

```
boxplot_131 <- ggplot(
  ml_data_131,
  aes(x = course_group, y = n_ml_areas, fill = course_group)
) +
  geom_boxplot() +
  labs(
    title = "ML-Related Interests by PSTAT 131 Enrollment",
    x = "Course Group",
    y = "Number of ML-Related Areas Selected"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

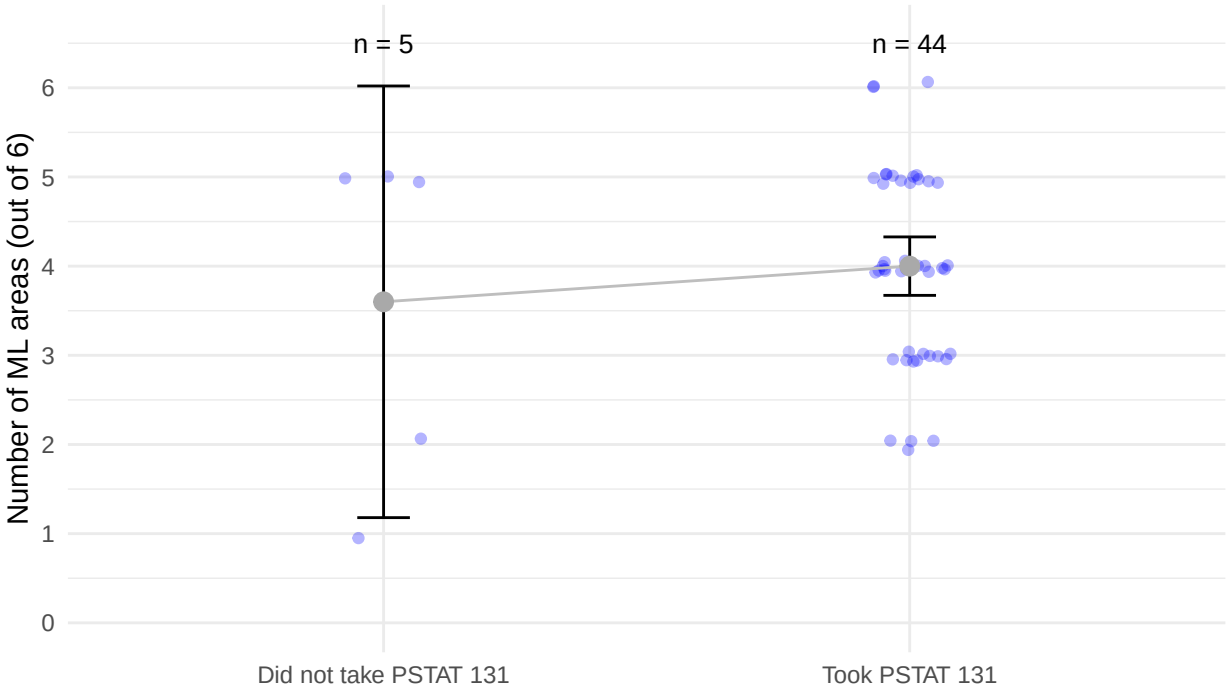
boxplot_131
```



Here we have a boxplot showing the distributions of each cohort. It appears that students who have not taken PSTAT 131 have a larger range, since there are only 5 students.

```
mean_131 <- ml_data_131 |>
  group_by(course_group) |>
  summarise(n = n(),
            mean_ml = mean(n_ml_areas),
            se = sd(n_ml_areas) / sqrt(n),
            lower = mean_ml - qt(0.975, n - 1) * se,
            upper = mean_ml + qt(0.975, n - 1) * se,
            .groups = 'drop')

ggplot(mean_131, aes(x = course_group, y = mean_ml)) +
  geom_jitter(data = ml_data_131, aes(y = n_ml_areas),
              width = 0.08, height = 0.08, alpha = 0.3, color = 'blue') +
  geom_line(aes(group = 1), color = 'gray') +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.1, color = 'black') +
  geom_point(size = 3, color = 'darkgray') +
  geom_text(aes(y = 6.5, label = paste0('n = ', n)), size = 3.5) +
  scale_y_continuous(limits = c(0, 6.6), breaks = 0:6) +
  labs(title = 'Mean number of ML areas selected, by PSTAT 131 enrollment',
       subtitle = 'Blue dots are group means; gray dots are individual students',
       caption = 'Bars are 95% confidence intervals for the mean.\nOnly 5 students did not take PSTAT 13
       x = NULL, y = 'Number of ML areas (out of 6)') +
  theme_minimal() +
  theme(plot.title.position = 'plot',
        plot.caption = element_text(hjust = 0, color = 'grey40'))
```

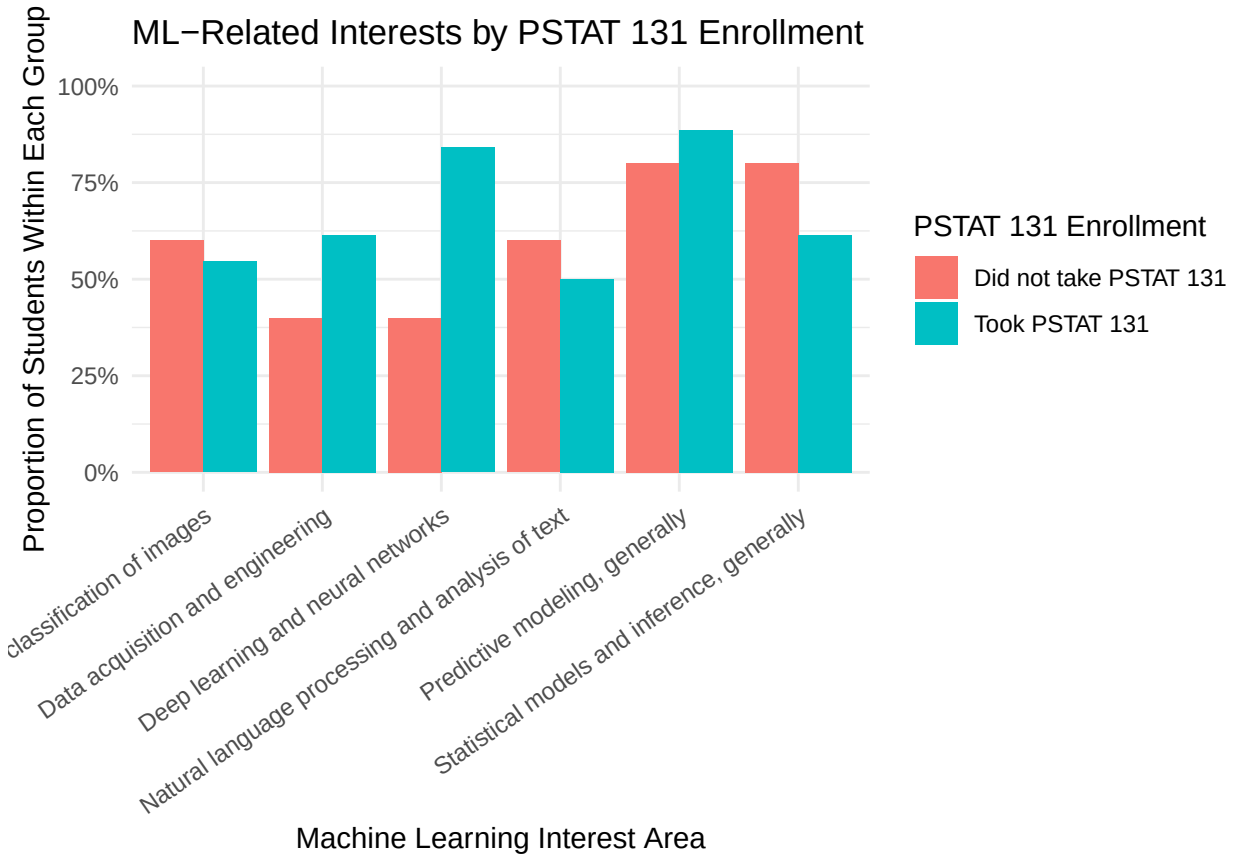



```r
area_proportions <- problem2 %>%
  mutate(
    took_131 = str_detect(
      coalesce(courses, ""),
      "\\bPSTAT 131\\b"
    ),
    areas = coalesce(areas, "")
  ) %>%
  select(response_id, took_131, areas) %>%
  crossing(area = ml_areas) %>%
  mutate(
    selected = str_detect(areas, fixed(area)),
    course_group = if_else(
      took_131,
      "Took PSTAT 131",
      "Did not take PSTAT 131"
    )
  ) %>%
  group_by(course_group, area) %>%
  summarise(
    n_selected = sum(selected),
    n_students = n_distinct(response_id),
    proportion = n_selected / n_students,
    .groups = "drop"
  )
```

```r
save_table(
  area_proportions,
  "ml_area_proportions_by_pstat131.csv"
)

area_chart <- ggplot(
  area_proportions,
  aes(x = area, y = proportion, fill = course_group)
) +
  geom_col(position = "dodge") +
  scale_y_continuous(
    labels = percent_format(),
    limits = c(0, 1)
  ) +
  labs(
    title = "ML-Related Interests by PSTAT 131 Enrollment",
    x = "Machine Learning Interest Area",
    y = "Proportion of Students Within Each Group",
    fill = "PSTAT 131 Enrollment"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 35,
      hjust = 1
    )
  )

area_chart
```




```
ggsave(
  file.path(fig_dir, "fig_2_ml_area_proportions.png"),
  plot = area_chart,
  width = 11,
  height = 7,
  dpi = 300
)
```

We can see that 'Deep learning and neural networks' has the biggest gap in proportion of students who find interest in that topic.

```
wilcox_result <- wilcox.test(
  n_ml_areas ~ course_group,
  data = ml_data_131,
  exact = FALSE
)

capture.output(
  wilcox_result,
  file = file.path(table_dir, "wilcoxon_test.txt")
)

wilcox_result
```

```
##
```

```
##  Wilcoxon rank sum test with continuity correction
##
## data:  n_ml_areas by course_group
## W = 107, p-value = 0.9319
## alternative hypothesis: true location shift is not equal to 0
```

We conducted a nonparametric test (particularly the Wilcoxon rank sum test) since the group of students who haven't taken PSTAT 131 is fairly small, at 5 students. This test is used to determine whether two independent samples come from populations with the same distribution or median. Since the p-value is high enough, we fail to reject the null hypothesis and conclude that there is no difference in the number of machine learning interests between students who have or have not taken PSTAT 131.

Takeaways from PSTAT 131

Overall, taking PSTAT 131 does not appear to be associated with electing more ML-related areas. The biggest gap showed up "Deep learning and neural networks", but the overall counts were similar and the Wilcoxon test found no significant difference. That said, this comparison is limited. With 44 of the 49 respondents having taken PSTAT 131, almost the whole class shares the same intro ML background, and the comparison group is only 5 students. This tells us that PSTAT 131 alone does not do much to separate students in our sample, so a more informative question is whether going beyond 131 makes a difference.

PSTAT 134 vs. non-PSTAT134

PSTAT 134 (Statistical Data Science) is a natural next step to look at. Since PSTAT 131 is a prerequisite, taking 134 reflects a student choosing to continue further into data science and machine learning, which might show up as a stronger or different ML interests. As done before, we can start by checking how many students fall into each group.

```
q134 <- problem2 |>
  mutate(group_134 = if_else(str_detect(courses, fixed('PSTAT 134')),
                             'Took PSTAT 134', 'Did not take PSTAT 134'))

q134 |>
  count(group_134, name = 'n_students') |>
  mutate(proportion = n_students / sum(n_students)) |>
  kable(digits = 3, caption = 'Students by PSTAT 134 enrollment')
```

Table 3: Students by PSTAT 134 enrollment

group_134	n_students	proportion
Did not take PSTAT 134	36	0.735
Took PSTAT 134	13	0.265

From this table, we can see that PSTAT 134 is not a commonly taken course among the students in our sample. Only 13 of the 49 respondents (26.5%) reported taking PSTAT 134, while 36 (73.5%) had not. This is a big contrast with PSTAT 131, which 44 of the 49 students (89.8%) had taken. Since PSTAT 131 is a prerequisite for PSTAT 134, most students have completed the introductory machine learning course but relatively few continued on to 134, leading to a unbalanced comparison.

Measuring ML Interest for Each Student

Each student could select any number of the eleven interest areas, so we need a single measure of how interested each student is in machine learning. To do this, we can check which of our six ML areas each student selected and count them, giving every student an ML interest score from 0 to 6. This allows us to compare both groups on the scale!

```
area_long_134 <- q134 |>
  crossing(area = ml_areas) |>
  mutate(selected = str_detect(areas, fixed(area)))

ml_data_134 <- area_long_134 |>
  group_by(response_id, group_134) |>
  summarise(n_ml_areas = sum(selected), .groups = 'drop')

head(ml_data_134)
```

```
## # A tibble: 6 x 3
##   response_id group_134              n_ml_areas
##         <dbl> <chr>                      <int>
## 1        1001 Did not take PSTAT 134         3
## 2        1002 Did not take PSTAT 134         5
## 3        1003 Did not take PSTAT 134         6
## 4        1004 Did not take PSTAT 134         2
## 5        1005 Did not take PSTAT 134         6
## 6        1006 Took PSTAT 134                 3
```

```
ml_data_134 |>
  count(group_134, n_ml_areas) |>
  pivot_wider(names_from = n_ml_areas, values_from = n, values_fill = 0) |>
  kable(caption = 'Number of students by ML areas selected (out of 6)')
```

Table 4: Number of students by ML areas selected (out of 6)

group_134	1	2	3	4	5	6
Did not take PSTAT 134	1	3	7	12	10	3
Took PSTAT 134	0	2	3	3	5	0

```
ml_data_134 |>
  group_by(group_134) |>
  summarise(n = n(),
            mean_ml_areas = mean(n_ml_areas),
            median_ml_areas = median(n_ml_areas),
            min = min(n_ml_areas),
            max = max(n_ml_areas)) |>
  kable(digits = 2, caption = 'Number of ML-related areas selected, by PSTAT 134 enrollment')
```

Summary

Table 5: Number of ML-related areas selected, by PSTAT 134 enrollment

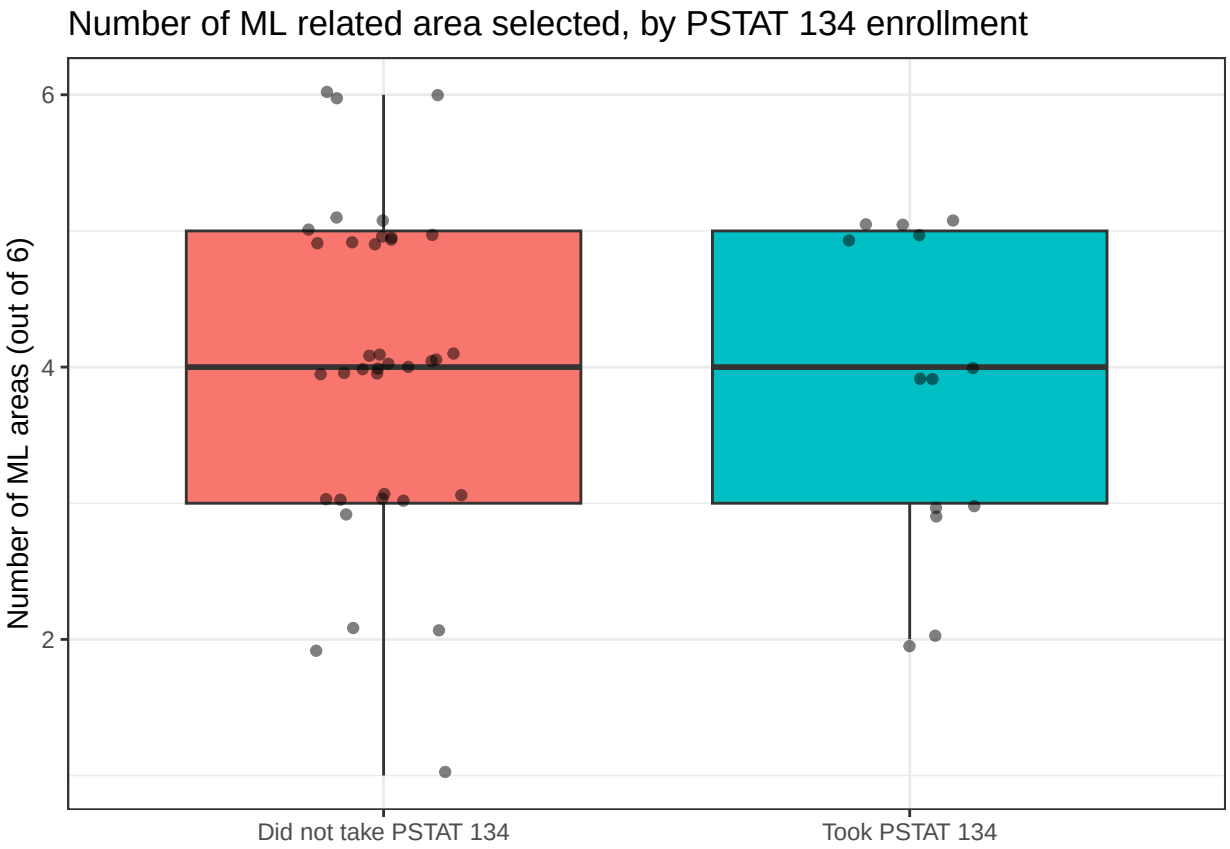
group_134	n	mean_ml_areas	median_ml_areas	min	max
Did not take PSTAT 134	36	4.00	4	1	6
Took PSTAT 134	13	3.85	4	2	5

Students who did not take PSTAT 134 selected slightly more ML related areas on average (4.00) than those who took PSTAT 134 (3.85), although both groups share the same median of 4. Something to note is that students who did not take PSTAT 134 also had a wider range (1 to 6 areas) than those who took it to (2 to 5 areas), likely because that group is nearly three times larger. In all, we can see that the two groups show very similar levels of ML interest, suggesting that taking PSTAT 134 is not really associated with selecting more ML related areas.

Distribution of ML Interest by Group

While the summary table gives us averages, it does not show how the scores are spread out within each group. A boxplot lets us compare the middle and range of ML interest scores side by side, and adding the individual points shows each student, which is especially helpful since the PSTAT 134 group only has 13 students.

```
ggplot(ml_data_134, aes(x = group_134, y = n_ml_areas, fill = group_134)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.15, height = 0.1, alpha = 0.5) +
  labs(title = "Number of ML related area selected, by PSTAT 134 enrollment",
       x = NULL, y = 'Number of ML areas (out of 6)') +
  theme_bw() +
  theme(legend.position = 'none')
```



The box plots for both groups look nearly identical: each has a median of 4 ML areas, and the middle 50% of students in both groups selected between 3 and 5 areas. As mention before the main difference is in the range, since students who did not take PSTAT 134 spread from 1 to 6 areas, while those who took PSTAT 134 stayed between 2 and 5. Overall, the plot supports our summary table, showing that the two groups have very similar distributions of ML interest.

ML Interest by Area

So far we have only looked at the total number of ML areas each student selected, which can hide differences in which areas each group prefers. Next, we can break the interest down by individual area and compare the proportion of students in each group who selected it. This shows whether students who took PSTAT 134 lean toward certain ML topics more than others.

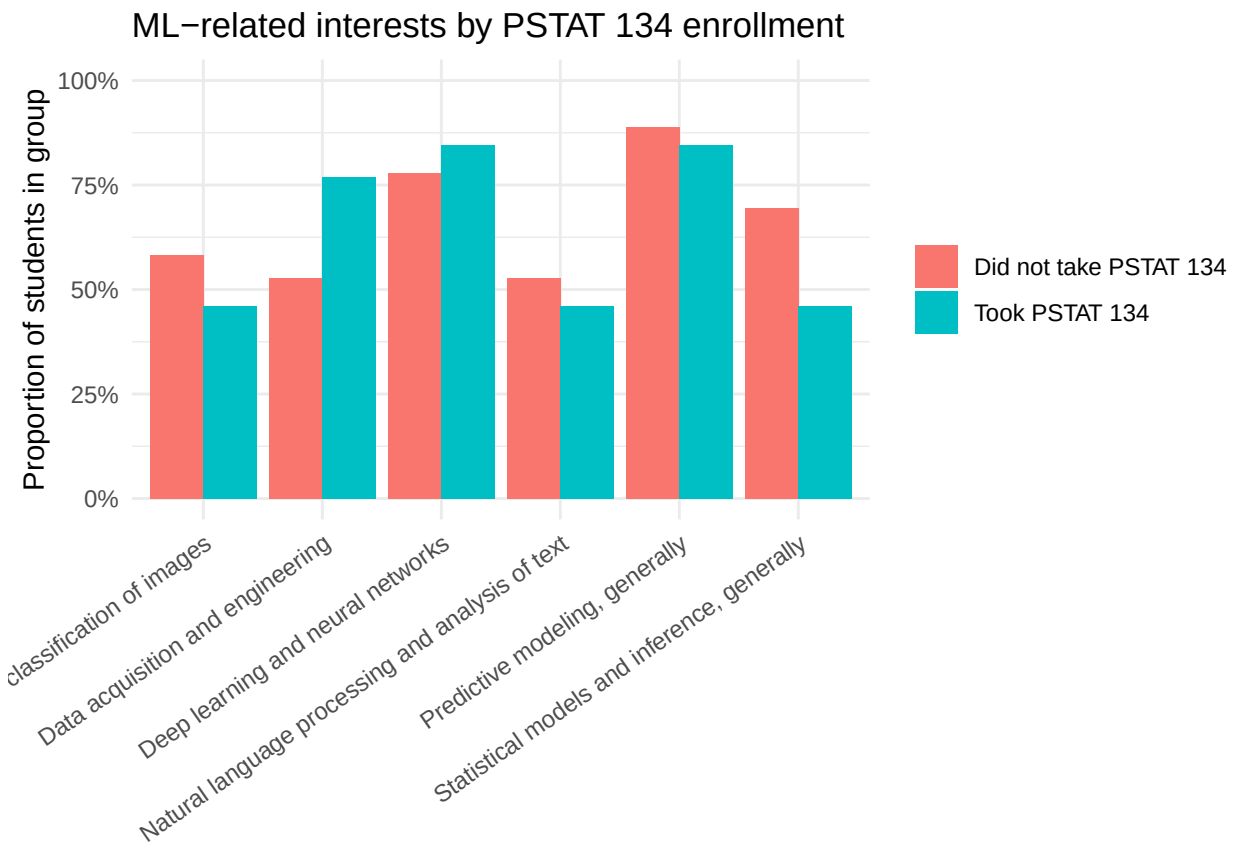
```
area_props_134 <- area_long_134 |>
  group_by(group_134, area) |>
  summarise(n_selected = sum(selected),
            n_students = n(),
            proportion = n_selected / n_students,
            .groups = 'drop')

area_props_134 |>
  select(area, group_134, proportion) |>
  pivot_wider(names_from = group_134, values_from = proportion) |>
  kable(digits = 3, caption = 'Proportion of students selecting each ML area, by PSTAT 134 enrollment')
```

Table 6: Proportion of students selecting each ML area, by PSTAT 134 enrollment

area	Did not take PSTAT 134	Took PSTAT 134
Analysis or classification of images	0.583	0.462
Data acquisition and engineering	0.528	0.769
Deep learning and neural networks	0.778	0.846
Natural language processing and analysis of text	0.528	0.462
Predictive modeling, generally	0.889	0.846
Statistical models and inference, generally	0.694	0.462

```
ggplot(area_props_134, aes(x = area, y = proportion, fill = group_134)) +
  geom_col(position = 'dodge') +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
  labs(title = 'ML-related interests by PSTAT 134 enrollment',
       x = NULL, y = 'Proportion of students in group', fill = NULL) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))
```



When we broke the ML interest down by area, some differences show up that the overall totals didn't catch initially. Students who took PSTAT 134 were a lot more into data acquisition and engineering (76.9% vs. 52.8%) and a little into deep learning (84.6% vs. 77.8%). On the other other hand, they picked statistical inference (46.2% vs. 69.4%). image analysis (46.2 % vs. 58.3%), and NLP (46.2% vs. 52.8%) less often. Predictive modeling was the top pick for both groups, with around 88.9% of non-134 students and 84/6% of 134 students choosing it. From this we can say that PSTAT 134 students might lean more toward data

engineering and away from traditional stats, but since there are only 13 of them, these differences could just be a random chance.

Testing for Differences

The tables and plots show some difference between the two groups, but with only 13 students in the PSTAT 134 group, we should check whether these gaps are actually meaningful or could just be random noise. To check, we use Fisher's exact test for each area, since it works well with small groups, and a Wilcoxon rank-sum test to compare the total number of ML areas picked, since those scores are counts rather than normally distributed values.

```
# Fisher's exact test

area_long_134 |>
  group_by(area) |>
  summarise(took_134 = mean(selected[group_134 == 'Took PSTAT 134']),
            not_134 = mean(selected[group_134 == 'Did not take PSTAT 134']),
            p_value = fisher.test(table(group_134, selected))$p.value) |>
  kable(digits = 3, caption = "Fisher's exact test by ML area")
```

Table 7: Fisher's exact test by ML area

area	took_134	not_134	p_value
Analysis or classification of images	0.462	0.583	0.525
Data acquisition and engineering	0.769	0.528	0.191
Deep learning and neural networks	0.846	0.778	0.709
Natural language processing and analysis of text	0.462	0.528	0.754
Predictive modeling, generally	0.846	0.889	0.650
Statistical models and inference, generally	0.462	0.694	0.184

```
# Wiclxo rank sum test
wilcox.test(n_ml_areas ~ group_134,
            data = ml_data_134,
            exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  n_ml_areas by group_134
## W = 249.5, p-value = 0.7251
## alternative hypothesis: true location shift is not equal to 0
```

From the Fisher's test, it seems like none of the six ML areas showed a statistically significant difference between the two groups, since every p value was well above 0.05. Even with the biggest gaps, like data engineering ($p = 0.191$) and statistical inference ($p = 0.184$), were not strong enough to rule out random chance. Additionally, the Wilcox rank sum test found no significant difference in the total of number of ML areas picked between the two groups ($W = 249.5$, $p = 0.725$). Meaning that students who took PSTAT 134 and those who didn't showed pretty much the same overall level of interest in machine learning.

Takeaways from PSTAT 134

There a few things to consider when looking at these results. First, the groups are pretty unbalanced, with only 13 students who took PSTAT 134 compared to 36 who didn't. That small group means one or two students can skew the percentage a lot, and our tests don't have much power to pick up a real difference. Second, the "did not take PSTAT 134" group isn't all the same, since it mixes 31 students who took PSTAT 131 with 5 who took neither course, so it isn't a clean comparison. The survey also let students pick as many areas as they wanted, our six ML areas was more a judgment call, also the data is self reported interest rather than actual skill, and since students choose their own courses, we cant determine that PSTAT 134 caused any differences.

PSTAT 134 + 131 vs. PSTAT 131 Only

Now we will answer our main question: among students who have taken PSTAT 131, do those who also took PSTAT 134 report more interest in machine learning? One issue with the PSTAT 134 vs. non-134 comparison is that the "did not take PSTAT 134" group also included the 5 students who never took PSTAT 131. To get a cleaner comparison, we can now look only at students who took PSTAT 131 and split them into those who stopped at 131 and those who went on to take 134 as well. This way, everyone has the same intro ML background, and we can see whether taking 134 on top of 131 makes a difference.

Filtering to PSTAT 131 students

```
area_131 <- area_long_134 |>
  filter(str_detect(courses, fixed('PSTAT 131'))) |>
  mutate(cohort = if_else(group_134 == 'Took PSTAT 134', '131 + 134', '131 only'))

ml_131 <- area_131 |>
  group_by(response_id, cohort) |>
  summarise(n_ml_areas = sum(selected), .groups = 'drop')

ml_131 |>
  count(cohort, name = 'n_students') |>
  kable(caption = 'Students by cohort')
```

Table 8: Students by cohort

cohort	n_students
131 + 134	13
131 only	31

This leaves us with 44 students: 31 who only took PSTAT 131 and 13 who took both.

Summary

```
comparison_summary <- ml_131 |>
  group_by(cohort) |>
  summarise(
```

```
    n = n(),
    mean_ml_areas = mean(n_ml_areas),
    median_ml_areas = median(n_ml_areas),
    .groups = "drop"
  )

kable(
  comparison_summary,
  digits = 3,
  caption = "ML Interests: PSTAT 131 Only vs Both Courses"
)
```

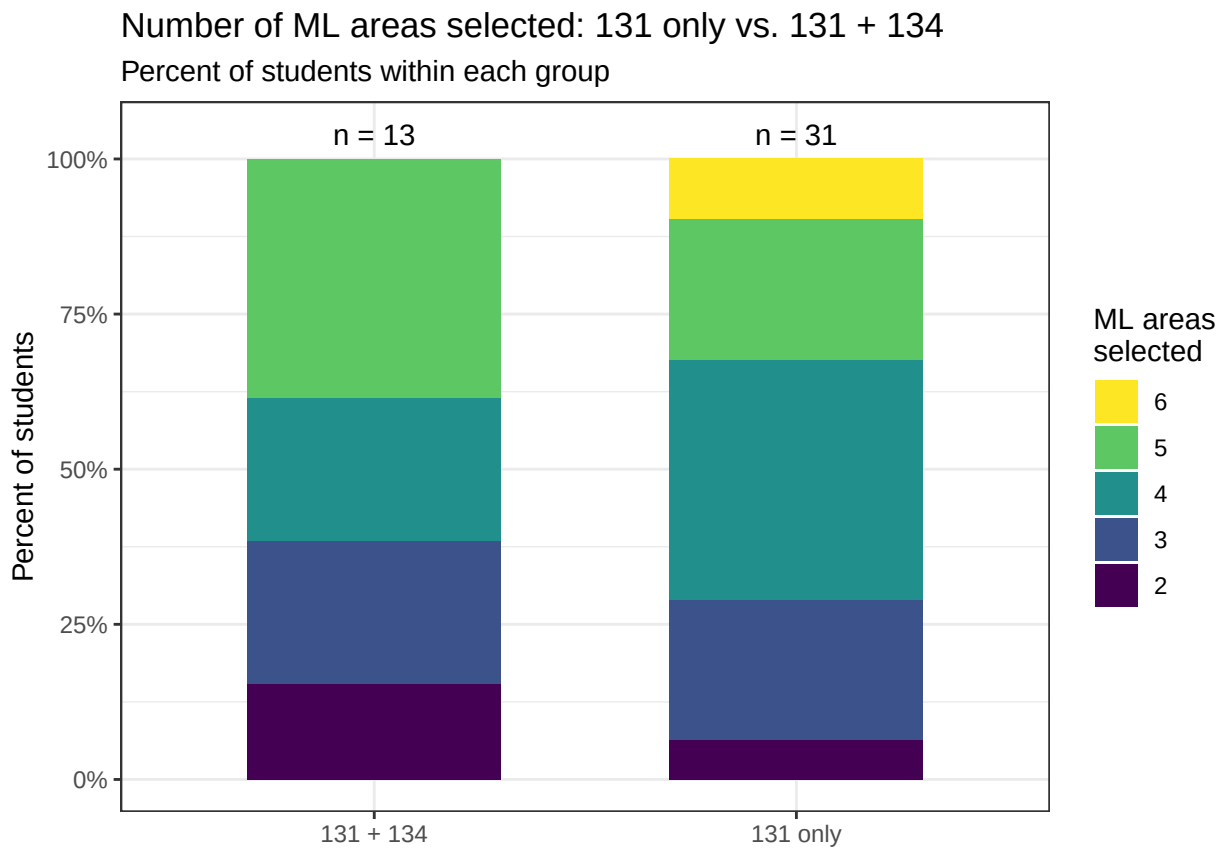
Table 9: ML Interests: PSTAT 131 Only vs Both Courses

cohort	n	mean_ml_areas	median_ml_areas
131 + 134	13	3.846	4
131 only	31	4.065	4

Student who took only PSTAT 131 picked slightly more ML areas on average (4.06 vs. 3.85), and both groups had a median of 4. These numbers are almost the same as in the PSTAT 134 vs. non-134 comparison, so removing the 5 students who never took 131 did not change much.

```
cohort_distribution <- ml_131 |>
  count(cohort, n_ml_areas) |>
  group_by(cohort) |>
  mutate(pct = n / sum(n)) |>
  ungroup() |>
  mutate(n_ml_areas = factor(n_ml_areas, levels = 6:1)) |>
  ggplot(aes(x = cohort, y = pct, fill = n_ml_areas)) +
  geom_col(width = 0.6) +
  geom_text(data = ml_131 |>
              count(cohort),
            aes(x = cohort, y = 1.04, label = paste0('n = ', n)),
            inherit.aes = FALSE) +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_viridis_d(direction = -1) +
  labs(title = 'Number of ML areas selected: 131 only vs. 131 + 134',
       subtitle = 'Percent of students within each group',
       x = NULL, y = 'Percent of students', fill = 'ML areas\nselected') +
  theme_bw()

cohort_distribution
```



The distributions for both groups look very similar, with most students in each group selecting between 3 and 5 ML areas. The main difference is at the top of the scale, where some students who only took PSTAT 131 selected all 6 areas, while none of the students who took both courses did.

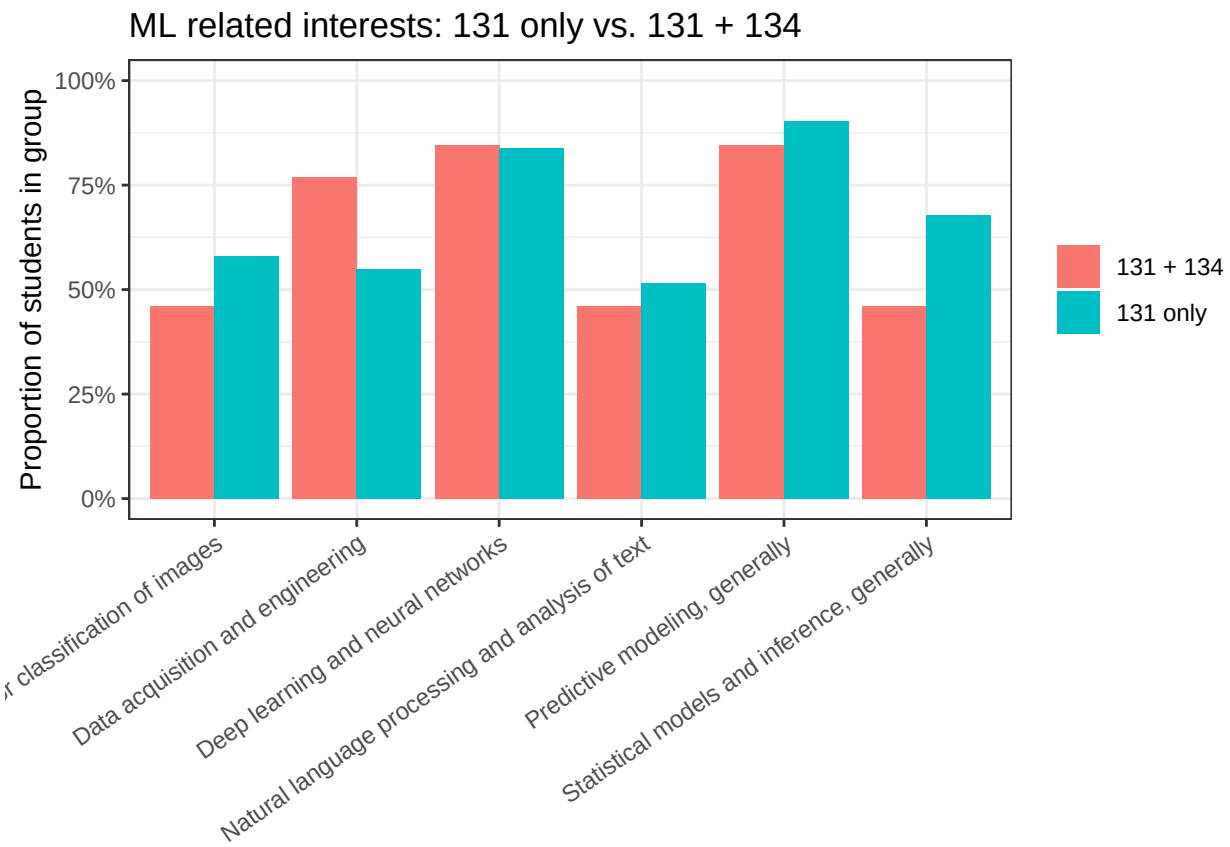
ML Interest by Area

```
comparison_proportions <- area_131 |>
  group_by(cohort, area) |>
  summarise(n_selected = sum(selected),
            n_students = n(),
            proportion = n_selected / n_students,
            .groups = 'drop')

save_table(comparison_proportions, 'ml_area_proportions_131_vs_131_134.csv')

comparison_plot <- ggplot(comparison_proportions,
                          aes(x = area, y = proportion, fill = cohort)) +
  geom_col(position = 'dodge') +
  scale_y_continuous(labels = scales::percent, limits = c(0,1)) +
  labs(title = 'ML related interests: 131 only vs. 131 + 134',
       x = NULL, y = 'Proportion of students in group', fill = NULL) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))

comparison_plot
```




```
ggsave(
  file.path(fig_dir, "fig_4_ml_area_proportions_131_vs_131_134.png"),
  plot = comparison_plot,
  width = 11,
  height = 7,
  dpi = 300
)
```

This follows from the same pattern as the PSTAT 134 vs. non-134 comparison. Students who took both courses leaned more toward data acquistiion and engineering (77% vs. 55%) and less toward statistical inference (46% vs. 68%). The other areas were fairly similar between the two groups.

Testing for Differences

Finally, we can check whether any of these differences are statistically significant. For each ML area, we use Fisher's exact test, since it works well with small groups like our 13 students who took both courses. The test also gives an odds ratio, which tells us how much more (or less) likely 131 + 134 students were to select each area compared to 131 only students.

```
fisher_131 <- area_131 |>
  group_by(area) |>
  summarise(test = list(fisher.test(table(
    factor(cohort, levels = c('131 only', '131 + 134')),
    selected))), .groups = 'drop') |>
  mutate(p_value = map_dbl(test, ~ .x$p.value),
```

```
        odds_ratio = map_dbl(test, ~ .x$estimate),
        lower = map_dbl(test, ~ .x$conf.int[1]),
        upper = map_dbl(test, ~ .x$conf.int[2])) |>
  select(-test)

save_table(fisher_131, 'fisher_tests_131_vs_131_134.csv')

kable(fisher_131,
      digits = 3,
      captions = "Fisher's exact test by ML area: 131 only vs. 131 + 134")
```

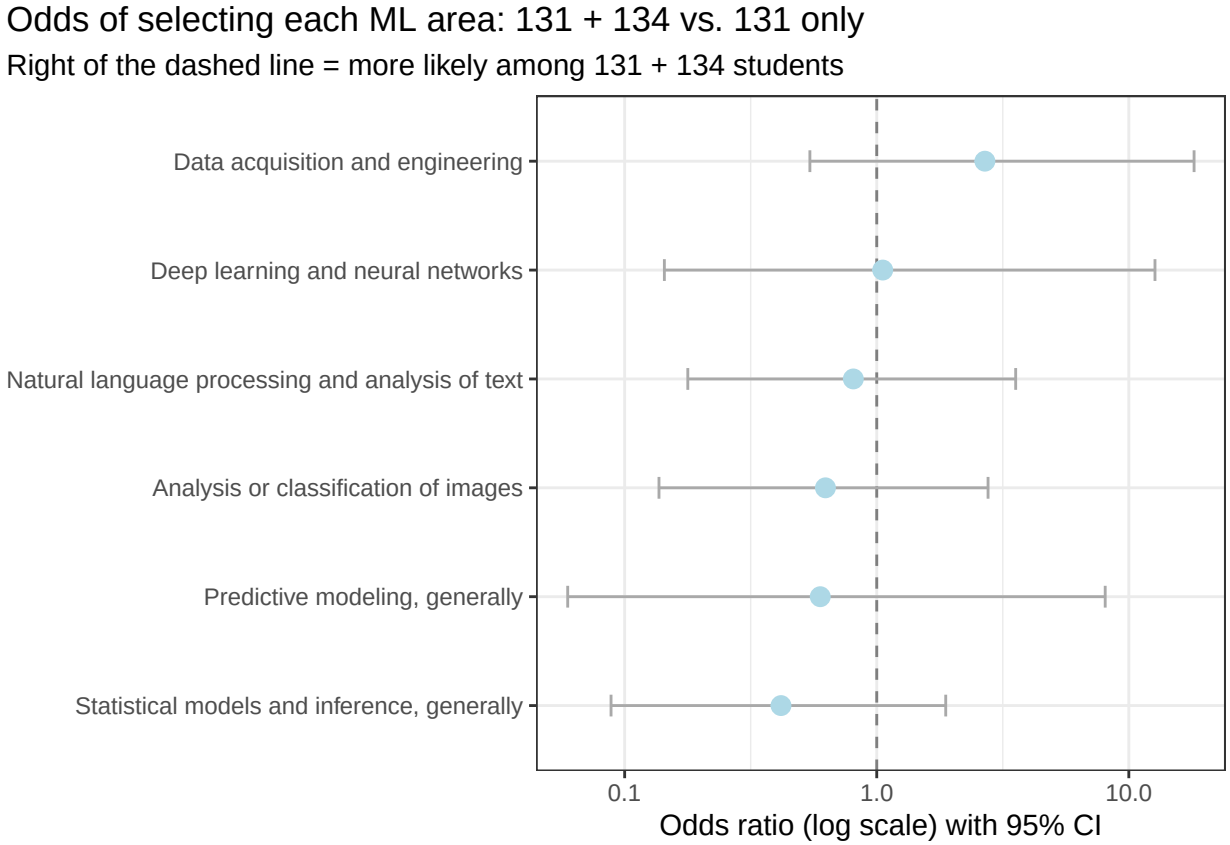
area	p_value	odds_ratio	lower	upper
Analysis or classification of images	0.522	0.626	0.137	2.764
Data acquisition and engineering	0.198	2.685	0.543	18.148
Deep learning and neural networks	1.000	1.056	0.144	12.700
Natural language processing and analysis of text	1.000	0.808	0.178	3.558
Predictive modeling, generally	0.623	0.597	0.059	8.059
Statistical models and inference, generally	0.309	0.417	0.088	1.878

For every ML area, the p-value is well above 0.05, so none of the differences between the two groups are statistically significant. The largest gap was in the data acquisition and engineering, where students who took both courses had about 2.7 times the odds of selecting it compared to 131 only students ($p = 0.198$). In the other direction, they had the lower odds of selecting statistical models and inference (odds ratio = 0.417, $p = 0.309$). Deep learning and NLP were almost identical between the groups, with odds ratios close to 1 and p-values of 1.. Every 95% confidence interval also includes 1, and more are very wide (data engineering was ranging from 0.54 to 18.15). What this means is that our data is consistent with taking PSTAT 134 having no effect at all, and the small 131 + 134 group having 13 students makes it hard to find any real difference.

```
ggplot(fisher_131, aes(x = odds_ratio, y = reorder(area, odds_ratio))) +
  geom_vline(xintercept = 1, linetype = 'dashed', color = 'grey50') +
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.2, color = 'darkgrey') +
  geom_point(size = 3, color = 'lightblue') +
  scale_x_log10() +
  labs(title = 'Odds of selecting each ML area: 131 + 134 vs. 131 only',
       subtitle = 'Right of the dashed line = more likely among 131 + 134 students',
       x = 'Odds ratio (log scale) with 95% CI', y = NULL) +
  theme_bw() +
  theme(plot.title.position = 'plot')
```

```
## Warning: `geom_errorbarh()` was deprecated in ggplot2 4.0.0.
## i Please use the `orientation` argument of `geom_errorbar()` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## `height` was translated to `width`.
```



This odds ratio plot shows the same results visually. The points for data engineering and deep learning fall to the right of the dashed line, while the other four areas fall to the left. However, every confidence interval crosses the dashed line at 1, so none of these differences can be separated from random chance.

Next, we compare the total number of ML areas each student selected. To achieve that, we can use a one sided Wilcoxon rank-sum test because the totals are small counts with only 13 students in one group, and our question asks whether 131 + 134 students report greater ML interest.

```
test_result <- wilcox.test(
  ml_131$n_ml_areas[ml_131$cohort == '131 + 134'],
  ml_131$n_ml_areas[ml_131$cohort == '131 only'],
  alternative = 'greater',
  exact = FALSE
)

test_result
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  ml_131$n_ml_areas[ml_131$cohort == "131 + 134"] and ml_131$n_ml_areas[ml_131$cohort == "131 o
## W = 186, p-value = 0.6654
## alternative hypothesis: true location shift is greater than 0
```

From this test it found no evidence that students who took both courses selected more ML areas than students who took only PSTAT 131 (W = 186, p = 0.665). In fact, their mean was slightly lower (3.85 vs. 4.06).

Together with the Fishers tests, this suggests that among students who have already taken PSTAT 131, also taking PSTAT 134 is not associated with greater interest in machine learning.

Drawbacks

Some drawbacks about this research question is that the survey allowed the respondent as many choices they wanted for area interests. In other words, someone could have checked all the boxes and make the cohorts even harder to determine. Ideally, we would have each person only selected one topic they are most interested in.

Also, since there were multiple questions on the survey, with the interest question being near the end, many students may have felt fatigued and not thinking as deep for their true answers.